RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251801040-2025-AIDS

B2

Started on	Thursday, 27 August 2026, 11:29 AM
State	Finished
Completed on	Thursday, 27 August 2026, 11:30 AM
Time taken	1 min 9 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int findFloor(int* nums, int low, int high, int x, int floor_val) {
    if (low > high) {
        return floor_val;
    }
    int mid = low + (high - low) / 2;
    if (nums[mid] == x) {
        return nums[mid];
    }
    else if (nums[mid] < x) {
        return findFloor(nums, mid + 1, high, x, nums[mid]);
    }
    else {
        return findFloor(nums, low, mid - 1, x, floor_val);
    }
}
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int nums[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &nums[i]);
    }
    int x;
    scanf("%d", &x);
    int floor_result = -1;
    floor_result = findFloor(nums, 0, n - 1, x, floor_result);
    printf("%d\n", floor_result);
    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔

	Input	Expected	Got	
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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